Department of
Computer Science and Engineering

University of Dhaka

Title: Implementation of the False Position, Newton–Raphson, and Secant Methods for finding roots of nonlinear equations

CSE 3212: Numerical Analysis Lab

Batch: 28/3rd Year 2nd Semester 2024

1 Objective(s)

- To understand the principles of the False Position, Newton–Raphson, and Secant methods.
- To implement the iterative algorithms in a computer program.
- To approximate the root of a nonlinear function to a desired accuracy.
- To compare convergence behavior and limitations of these methods.

2 Theory

2.1 False Position (Regula Falsi) Method

The False Position method is an improvement of the Bisection method. Like Bisection, it requires an initial interval $[a, b]$ such that $f(a) \cdot f(b) < 0$. Instead of halving the interval, the method uses a straight line between $(a, f(a))$ and $(b, f(b))$ and takes the x-intercept as the new approximation:

$$c = \frac{af(b) - bf(a)}{f(b) - f(a)}.$$

The process continues by updating either a or b, depending on the sign of $f(c)$.

2.2 Newton–Raphson Method

The Newton–Raphson method uses tangents to approximate the root. Starting with an initial guess x_0, the next approximation is given by:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}.$$

This method converges quadratically near the root, provided $f'(x) \neq 0$ and the initial guess is sufficiently close to the root.

2.3 Secant Method

The Secant method is similar to Newton–Raphson, but does not require explicit derivative calculation. It approximates the derivative using two nearby points:

$$x_{i+1} = x_i - f(x_i) \cdot \frac{x_i - x_{i-1}}{f(x_i) - f(x_{i-1})}.$$

This method is useful when $f'(x)$ is difficult to compute.

3 Procedure

False Position

1. Choose $[a, b]$ such that $f(a) \cdot f(b) < 0$.
2. Compute
$$c = \frac{af(b) - bf(a)}{f(b) - f(a)}.$$
3. Check $f(c)$:
   - If $f(a) \cdot f(c) < 0$, set $b = c$.
   - Else set $a = c$.
4. Repeat until $|f(c)|$ or $|b - a|$ is less than tolerance.

Newton–Raphson

1. Choose initial guess x_0.
2. Compute
$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}.$$
3. Repeat until $|x_{i+1} - x_i|$ or $|f(x_{i+1})|$ is less than tolerance.

Secant

1. Choose two initial guesses x_0 and x_1.
2. Compute
$$x_{i+1} = x_i - f(x_i) \cdot \frac{x_i - x_{i-1}}{f(x_i) - f(x_{i-1})}.$$
3. Repeat until $|x_{i+1} - x_i|$ is less than tolerance.

4 Example Execution (Detailed Calculations)

We use the test function
$$f(x) = x^3 - x - 2,$$
whose real root is approximately $x \approx 1.5213797$. Below we show detailed iterative calculations for the three methods.

4.1 False Position (Regula Falsi)

Given: interval $[a, b] = [1, 2]$, so $f(1) = -2$, $f(2) = 4$ (sign change holds).

The false position formula:
$$c = \frac{a\,f(b) - b\,f(a)}{f(b) - f(a)}.$$

1. **Iteration 1:**
$$c_1 = \frac{1 \cdot 4 - 2 \cdot (-2)}{4 - (-2)} = \frac{4 + 4}{6} = \frac{8}{6} = 1.333333\ .$$
Evaluate:
$$f(c_1) = (1.333333)^3 - 1.333333 - 2 \approx 2.370370 - 1.333333 - 2 = -0.962963.$$
Since $f(c_1) \cdot f(2) < 0$? No ($f(c_1) < 0$, $f(2) > 0$), so new interval becomes $[c_1, 2] = [1.333333, 2]$.
2. **Iteration 2:**
$$a = 1.333333,\ b = 2,\ f(a) \approx -0.962963,\ f(b) = 4.$$
$$c_2 = \frac{1.333333 \cdot 4 - 2 \cdot (-0.962963)}{4 - (-0.962963)} = \frac{5.333332 + 1.925926}{4.962963} \approx \frac{7.259258}{4.962963} \approx 1.462687.$$
$$f(c_2) \approx (1.462687)^3 - 1.462687 - 2 \approx 3.130005 - 1.462687 - 2 = -0.332682.$$
Root is in $[1.462687, 2]$.
3. **Iteration 3:**
$$a = 1.462687,\ b = 2,\ f(a) \approx -0.332682,\ f(b) = 4.$$
$$c_3 = \frac{1.462687 \cdot 4 - 2 \cdot (-0.332682)}{4 - (-0.332682)} \approx \frac{5.850748 + 0.665364}{4.332682} \approx \frac{6.516112}{4.332682} \approx 1.503341.$$
$$f(c_3) \approx (1.503341)^3 - 1.503341 - 2 \approx 3.398 - 1.503341 - 2 \approx -0.10534.$$
Root in $[1.503341, 2]$.

4. **Iteration 4:**

$$a = 1.503341,\ b = 2,\ f(a) \approx -0.10534,\ f(b) = 4.$$

$$c_4 = \frac{1.503341 \cdot 4 - 2 \cdot (-0.10534)}{4 - (-0.10534)} \approx \frac{6.013364 + 0.21068}{4.10534} \approx \frac{6.224044}{4.10534} \approx 1.516847.$$

$$f(c_4) \approx (1.516847)^3 - 1.516847 - 2 \approx -0.03801.$$

Root in $[1.516847, 2]$.

5. **Iteration 5:**

$$a = 1.516847,\ b = 2,\ f(a) \approx -0.03801,\ f(b) = 4.$$

$$c_5 = \frac{1.516847 \cdot 4 - 2 \cdot (-0.03801)}{4 - (-0.03801)} \approx \frac{6.067388 + 0.07602}{4.03801} \approx \frac{6.143408}{4.03801} \approx 1.520784.$$

$$f(c_5) \approx (1.520784)^3 - 1.520784 - 2 \approx -0.01356.$$

Root in $[1.520784, 2]$.

6. **Iteration 6:**

$$a = 1.520784,\ b = 2,\ f(a) \approx -0.01356,\ f(b) = 4.$$

$$c_6 \approx 1.521356 \quad \text{(after substitution and division)},$$

$$f(c_6) \approx -0.00100.$$

After a few more iterations (not shown) the method converges to

$$c \approx 1.52138.$$

Remarks: False Position often converges faster than Bisection, but if one endpoint's $|f|$ remains much larger, convergence may slow (stagnation of one end).

4.2 Newton–Raphson Method

Given: initial guess $x_0 = 1.5$. For $f(x) = x^3 - x - 2$ we have derivative

$$f'(x) = 3x^2 - 1.$$

Iteration formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

1. **Iteration 0 → 1:**

$$f(1.5) = (1.5)^3 - 1.5 - 2 = 3.375 - 1.5 - 2 = -0.125.$$

$$f'(1.5) = 3(1.5)^2 - 1 = 3(2.25) - 1 = 6.75 - 1 = 5.75.$$

$$x_1 = 1.5 - \frac{-0.125}{5.75} = 1.5 + 0.02173913 \approx 1.52173913.$$

2. **Iteration 1 → 2:** Evaluate at x_1:

$$f(1.52173913) \approx (1.52173913)^3 - 1.52173913 - 2 \approx 0.003806.$$

$$f'(1.52173913) \approx 3(1.52173913)^2 - 1 \approx 3(2.31669) - 1 \approx 5.95007.$$

$$x_2 = 1.52173913 - \frac{0.003806}{5.95007} \approx 1.52137971.$$

3. **Iteration 2 → 3:** Evaluate at x_2:

$$f(1.52137971) \approx (1.52137971)^3 - 1.52137971 - 2 \approx 1.0 \times 10^{-6} \text{ (very small)}.$$

$$x_3 \approx 1.5213797 \quad \text{(converged to machine accuracy)}.$$

Newton–Raphson converged very rapidly (quadratic convergence) from $x_0 = 1.5$ to the root $x \approx 1.5213797$ in just 2–3 iterations.

4.3 Secant Method

Given: two initial guesses $x_0 = 1$, $x_1 = 2$.

Secant formula:

$$x_{n+1} = x_n - f(x_n)\,\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}.$$

1. **Iteration 0 → 2:**

$$f(1) = -2, \qquad f(2) = 4.$$

$$x_2 = 2 - 4 \cdot \frac{2-1}{4-(-2)} = 2 - \frac{4}{6} = 2 - 0.6666667 = 1.3333333.$$

$$f(1.3333333) \approx -0.962963.$$

2. **Iteration 1 → 3:** Use $x_1 = 2$, $x_2 = 1.3333333$:

$$x_3 = 1.3333333 - (-0.962963) \cdot \frac{1.3333333 - 2}{-0.962963 - 4} \approx 1.3333333 - (-0.962963) \cdot \frac{-0.6666667}{-4.962963}$$

$$\approx 1.3333333 - 0.1299917 \approx 1.4623416.$$

$$f(1.4623416) \approx -0.3343.$$

3. **Iteration 2 → 4:** Use $x_2 = 1.3333333$, $x_3 = 1.4623416$:

$$x_4 = 1.4623416 - (-0.3343) \cdot \frac{1.4623416 - 1.3333333}{-0.3343 - (-0.962963)} \approx 1.4623416 - (-0.3343) \cdot \frac{0.1290083}{0.628663}$$

$$\approx 1.4623416 + 0.0687494 \approx 1.5310910.$$

$$f(1.5310910) \approx 0.0611.$$

4. **Iteration 3 → 5:** Use $x_3 = 1.4623416$, $x_4 = 1.5310910$:

$$x_5 = 1.5310910 - 0.0611 \cdot \frac{1.5310910 - 1.4623416}{0.0611 - (-0.3343)} \approx 1.5310910 - 0.0611 \cdot \frac{0.0687494}{0.3954}$$

$$\approx 1.5310910 - 0.010552 \approx 1.520539.$$

$$f(1.520539) \approx -0.0049.$$

The secant method converges super-linearly and in this example homes in on the root within a few iterations; further iteration gives $x \approx 1.52138$.

Summary of Example Results

- False Position (after several iterations): $x \approx 1.52138$.
- Newton–Raphson (from $x_0 = 1.5$): converges to $x \approx 1.5213797$ in 2–3 iterations.
- Secant (from $x_0 = 1, x_1 = 2$): converges to $x \approx 1.52138$ in a few iterations.

Notes for students

1. All numerical values above are *approximate*; please compute with a calculator or a small program for more digits. Show your intermediate steps (function values, derivatives, numerator/denominator in formulas) clearly in the lab report.
2. Report and tabulate: iteration number, current approximations, f at those approximations, new approximation, and approximate relative error:

$$e_a(\%) = \left| \frac{x_k - x_{k-1}}{x_k} \right| \times 100\%.$$

3. For Newton–Raphson, always display $f(x_k)$ and $f'(x_k)$. For False Position and Secant, display both bracket endpoints and the computed c or x_{k+1}.

5 Lab Tasks (Please implement yourself and show the output to the instructor)

5.1 Problem 1

The velocity y of a falling parachutist is given by

$$y = \frac{g\,m}{c}\Big(1 - e^{-(c/m)t}\Big),$$

where $g = 9.81$ m/s^2. For a parachutist with drag coefficient $c = 15$ kg/s, compute the mass m so that the velocity is $y = 36$ m/s at $t = 10$ s. Use the **False-Position method** to determine m to a level of approximate relative error $e_s = 0.001\%$. Employ initial guesses

$$x_\ell = 40\text{kg}, \qquad x_u = 80\text{kg}.$$

Report x_l, x_u, approximate root x_r, $f(x_r)$, and approximate error $\epsilon_a(\%)$ after each iterations in tabular format. To avoid round-off error, use 10 places after the decimal point. Also, generate a graph of approximate error $\epsilon_a(\%)$ vs the number of iterations to visualize the convergence of error.

5.2 Problem 2

Solve problem given in 5.1 using the **Newton–Raphson method** to determine m such that the approximate relative error $e_s \leq 0.001\%$. Assume initial guess $m_0 = 40$kg.

Report m_k, $f(m_k)$, $f'(m_k)$, and approximate error $\epsilon_a(\%)$ after each iterations in tabular format. To avoid round-off error, use 10 places after the decimal point. Also, generate a graph of approximate error $\epsilon_a(\%)$ vs the number of iterations to visualize the convergence of error.

5.3 Problem 3

Solve problem given in 5.1 using the **Secant method** to determine m such that the approximate relative error $e_s \leq 0.001\%$. Assume initial guess $m_0 = 30$kg and $m_1 = 40$kg.

Report m_k, $f(m_k)$, $f'(m_k)$, and approximate error $\epsilon_a(\%)$ after each iterations in tabular format. To avoid round-off error, use 10 places after the decimal point. Also, generate a graph of approximate error $\epsilon_a(\%)$ vs the number of iterations to visualize the convergence of error.

6 Lab Report (Please implement yourself and submit as a lab report)

A tank is draining through an orifice. The governing relation between height h (m) and time t (s) is nonlinear. For a given system, you want to find h such that flow balance holds:

$$f(h) = h^3 - 10h + 5e^{-h/2} - 2 = 0$$

Tasks for Students

1. Solve the problem using **Bisection** and **False-Position** methods with a suitable interval (e.g., $[0.1, 0.4]$) to determine h such that the approximate relative error $e_s \leq 0.001\%$. To avoid round-off error, use 5 places after the decimal point.
2. Apply the **Newton–Raphson** and **Secant** methods with initial guesses $h_0 = 1.5$ and $h_1 = 2.0$ to determine h such that the approximate relative error $e_s \leq 0.001\%$. To avoid round-off error, use 5 places after the decimal point.
3. Report all of the necessary data in tabular format.
4. Compare the convergence rates of all methods by plotting the error $\epsilon_a(\%)$ versus the number of iterations.

7 Policy

Copying from the Internet, classmates, seniors, or from any other source is strongly prohibited. 100% marks will be *deducted* if any such copying is detected.